Udara Kavishka Nalawansa

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Personal Statement

- First Class IT graduate and Associate Software Engineer working day to day in Python, building REST APIs with FastAPI and Flask on PostgreSQL behind Next.js and TypeScript frontends.
- Delivers features end to end across frontend, backend, and infrastructure, deploying with Docker and CI/CD pipelines and resolving live production issues.
- Final year research applied deep learning and continual learning to Kubernetes autoscaling.
- Works with a product mindset: asks what problem sits behind a request, weighs scope against what users and the business actually need, and stays with a feature from requirement through deployment and the support that follows. Seeking a software engineering or DevOps role.

Education

University of Vavuniya (former Vavuniya Campus of the University of Jaffna) 2022 – 2026
BSc Honours in Information Technology **First Class**

St Joseph's College, Colombo 10 2005 – 2020

Professional Experience

Xaventra (Private) Limited, Colombo Feb 2026 – Present
Associate Software Engineer

- Building and maintaining a company secretarial and business compliance platform, spanning a Flask admin CRM, a FastAPI backend, and a Next.js client portal and marketing site, shipping 288 merged pull requests and 37 database migrations in eight months.
- Delivered features end to end using Python, Flask, FastAPI, SQLAlchemy, PostgreSQL, and Alembic migrations, and automated deployments with GitHub Actions self-hosted runners monitored through Sentry.
- Designed and integrated RESTful APIs in FastAPI and PostgreSQL against a Next.js and TypeScript frontend, following OpenAPI-driven development so the generated client stays in sync with the backend contract.
- Diagnosed and resolved production issues through log analysis, database inspection, and API tracing, and supported CI/CD workflows, Alembic migrations, and incident response with root-cause analysis.

Core Competencies

Programming: Python, Java, JavaScript, TypeScript, Bash, Go

Frontend Technologies: React.js, Next.js, TypeScript, HTML, TailwindCSS, shadcn/ui, React Hook Form, Zod, TanStack Query, React Native

Backend Technologies: Node.js, Express.js, FastAPI, Flask, SQLAlchemy, Alembic, Pydantic, REST APIs, Python, Java

Databases: PostgreSQL, MySQL, MongoDB

DevOps & Cloud: Docker, Docker Compose, Podman, Kubernetes, Linux, AWS, GCP, Vercel

CI/CD & Automation: GitHub Actions (self-hosted runners), Jenkins, SonarQube, GitLab CLI, ArgoCD, Terraform, Ansible

Monitoring: Prometheus, Grafana, Sentry, ELK Stack

Tools: Git, GitHub, Claude Code (agentic coding), MCP, Postman, OpenAPI, Figma, Huly, Trello

Soft Skills: Leadership, Teamwork, Problem Solving, Adaptability, Presentation, Time Management

Projects

CeylonRoots (Link)
University of Vavuniya – 2025

- Built a travel agency platform using Next.js (frontend) and Spring Boot (backend).
- Designed to showcase Sri Lanka's cultural heritage while enabling travel itinerary customization, bookings, and customer support.
- Integrated responsive UI and backend APIs for seamless user interaction.

URL Shortener (Go) (Link)
Self Learned– 2025

- Developed a high-performance URL shortening service using Go, supporting URL mapping, redirection, and persistence.
- Designed for low-latency request handling and scalability.
- Implemented clean API design and efficient data handling for fast redirections.

Kubernetes Lab Cluster Setup (Blog Post)
University of Vavuniya – 2025

- Built and configured a 4-node physical Kubernetes cluster for hands-on orchestration and research experimentation.
- Deployed workloads, managed nodes, and monitored cluster behavior under varying resource loads.
- Established the cluster as a research platform for performance testing and adaptive CPU auto-scaling experiments.

Research Experience

Undergraduate Research (BSc Hons IT) 2025 – 2026

- Completed a final year research project on **adaptive CPU auto scaling using hybrid neural networks and continual learning within Kubernetes orchestration**.
- Developed a hybrid LSTM and MLP CPU demand forecasting model with continual learning through Elastic Weight Consolidation and Experience Replay, trained on Google Cluster Traces.
- Deployed and managed a controlled Kubernetes test environment and benchmarked the predictive autoscaler against the standard Horizontal Pod Autoscaler using a Prometheus based evidence pipeline.
- Analyzed cluster behavior and resource utilization patterns to improve scaling efficiency.

Activities

Christian Society, University of Vavuniya
President | Dec 2024 – 2025

Sri Lanka University Catholic Students' Movement (SLUCSM)
Committee Member (Representative of University of Vavuniya) | 2023 – 2024

IEEE Student Branch, University of Vavuniya
Member | 2023 – Present

Leo Club of University of Vavuniya
Chief Editor | 2023 – 2024

ITCS Information Technology and Computing Society
Member | 2023 – Present

University Colours in Basketball
Awarded for outstanding performance | 2024

References

Dr. S. Kirushanth
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Department of Physical Science
University of Vavuniya
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